Hood S. Mukiibi

Security Operations Engineering | Project Management

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Portfolio

EXPERIENCE

Carnegie Mellon University, PA — Systems *Security Operations Engineer*

JUNE 2023 - MARCH 2025

- **Built a cutting-edge Security Operations Center (SOC):** Led the design and deployment of an open-source SOC, leveraging Wazuh as the core SIEM and HIDS platform. Configured Wazuh for centralized log collection, file integrity monitoring, intrusion detection, and compliance reporting, significantly improving visibility across networked systems.
- **AI-Assisted Security Operations:** Integrated LLM-based AI assistants into SOC workflows alongside Wazuh to enhance alert triage, rule tuning, and case management. This fusion improved SOC analysts' efficiency and decision-making in real-time incident response.
- **Enhanced incident response capabilities:** Utilized Wazuh's agent-based monitoring and threat intelligence feeds to detect anomalies and spearhead forensic investigations. This reduced MTTD and MTTR by enabling timely and actionable insights.
- **Improved data security by 30%:** Developed and enforced data classification and protection policies in tandem with Wazuh's policy monitoring modules, ensuring compliance with ISO 27001, NIST, and GDPR standards.
- **Bridged academia and industry:** Designed SOC training programs using Wazuh to simulate real-world attack scenarios and monitoring techniques, preparing students for industry challenges.
- **Influenced cybersecurity policy:** Provided advisory support across sectors, informed by insights generated from Wazuh-based monitoring and compliance tools deployed in test environments.

Rohde and Schwarz, Leipzig, Germany — *Cybersecurity Research Engineer II (5G & IIOT-ICS Security)*

OCTOBER 2023 - JUNE 2024

- **Strengthened 5G network security:** Conducted in-depth security assessments, uncovering and mitigating vulnerabilities in containerized network functions.
- **Optimized security monitoring:** Designed and deployed Prometheus-based monitoring solutions, enhancing real-time

CERTIFICATIONS & TRAINING

ISC² Certified in Cybersecurity (CC)

ISC² Certified Information System Security Professional (CISSP) ~ Ongoing

Information Technology Infrastructure Library (ITIL)

Security+

SKILLS

Systems Administration | Security Operations (SIEM, EDR, XDR) | Threat Hunting and Incident Response | Penetration Testing and Vulnerability Assessment | Security Automation & SOAR | Cloud Security (Azure, AWS, GCP - CNAPP, CSPM) | Governance, Risk & Compliance (ISO27001, NIST, GDPR) | Cybersecurity Training and Mentorship.

TOOLS & TECHNOLOGIES

SIEM: Splunk | Elastic | QRadar | Sentinel | Wazuh
Virtualization: VMware | Hyper-V
EDR/XDR: CrowdStrike | Microsoft Defender | Elastic Security | Zeek
Cloud Security: AWS Security Hub | Azure Defender | GCP Chronicle | CNAPP

threat visibility and performance tracking.

- **Improved threat detection accuracy:** Implemented Deep Packet Inspection (DPI) solutions, leading to better anomaly detection and proactive security measures.
- **Conducted high-impact attack simulations:** Modeled targeted attacks on 5G core architectures, identifying security gaps and proposing mitigation strategies.
- **Secured IIoT environments:** Identified and addressed critical vulnerabilities, bolstering security for mission-critical industrial systems.

Tabiri Analytics Inc., Rwanda — *Sr. Security Consultant (Threat Detection & Cybersecurity Risk Management)*

DECEMBER 2018 - OCTOBER 2023

- **Optimized threat detection capabilities:** Deployed and fine-tuned SIEM solutions across multiple client environments, with Wazuh serving as a primary platform for several organizations. Custom rules and decoders were developed to improve threat detection accuracy and reduce false positives by over 30%.
- **Reduced incident response time by 40%:** Integrated Wazuh with orchestration tools and custom Python/PowerShell scripts to automate alert triage, log correlation, and response actions, significantly accelerating forensic investigations and containment.
- **Enhanced cloud security:** Leveraged Wazuh agents to monitor cloud workloads in AWS and Azure environments, ensuring visibility into file integrity, unauthorized access, and compliance misconfigurations in alignment with ISO 27001 and PCI-DSS standards.
- **Strengthened risk management:** Conducted enterprise-level vulnerability assessments and threat modeling using data collected through Wazuh's vulnerability detector and endpoint telemetry. This helped align cybersecurity strategies with NIST, CIS Controls, and industry best practices.
- **Influenced executive decision-making:** Delivered client-specific security reports and dashboards derived from Wazuh analytics to C-level executives, translating complex technical risks into actionable business insights.
- **Led cybersecurity training initiatives:** Trained internal SOC teams and client engineers on the effective use of Wazuh for real-time monitoring, alert handling, and compliance auditing—improving in-house detection and response capabilities across sectors including finance, healthcare, energy, and education.

Carnegie Mellon University, PA — *Graduate Teaching Assistant*

AUGUST 2017 - MAY 2023

- **Mentored next-generation cybersecurity professionals:** Provided hands-on guidance in key areas such as threat hunting, SIEM tools, and secure system design.

Offensive Security: Burp Suite | Metasploit | Wireshark
Automation: Python | PowerShell | Bash

- **Developed industry-relevant curriculum:** Assisted in course enhancement, integrating real-world cybersecurity challenges into academic training.
- **Facilitated hands-on labs:** Led technical workshops and interactive lab sessions, improving students' practical cybersecurity skills.

Rwanda Revenue Authority, Rwanda — *Jr. Information Security Consultant*

MAY 2017 - SEPTEMBER 2017

- **Identified and mitigated security gaps:** Conducted comprehensive security audits, leading to actionable security improvements.
- **Enhanced compliance readiness:** Assisted in ISO/IEC 27001 certification preparation by developing security policies and risk mitigation strategies.
- **Improved threat detection:** Implemented MITRE ATT&CK-based investigations, refining SOC operations and threat response protocols.

MTN, Uganda — *IT End User Support Lead*

OCTOBER 2015 - JULY 2016

- **Led cross-functional IT support teams:** Provided technical support across five OPCOs, maintaining SLA compliance and minimizing downtime.
- **Optimized ITSM processes:** Streamlined ticketing and incident management, improving service desk efficiency.
- **Implemented process improvements:** Identified and eliminated bottlenecks, enhancing overall IT service delivery.

MTN, Uganda — *Lead Service Desk Engineer*

JANUARY 2013 - SEPTEMBER 2015

- **Improved service desk operations:** Reduced ticket resolution times through workflow optimization and automation.
- **Enhanced SLA performance:** Identified process inefficiencies and implemented corrective measures, increasing service quality.
- **Led ITIL best practice adoption:** Standardized service delivery and incident response, improving IT governance.

EDUCATION

Carnegie Mellon University, Pittsburgh, PA — *Master of Science in Information Technology*

AUGUST 2016 - MAY 2018

Makerere University, Uganda — *Bachelor of Science in Computer Science*

AUGUST 2008 - JANUARY 2012